Andrew Kerfoot

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PROFILE

Ecoinformatics graduate (GPA 3.94) with hands-on experience building Python and R workflows for ecological disease modeling, probabilistic forecasting, and climate-data integration. Seeking data, research, software, or environmental analytics roles where reproducible quantitative analysis, scientific communication, and applied programming skills add direct value.

EDUCATION

B.S. Informatics — Ecology Emphasis | Northern Arizona University, Flagstaff AZ *May 2026*

GPA: 3.94 | Study Abroad: University of Liverpool, UK (Fall 2025)

TECHNICAL SKILLS

Programming: Python (Advanced) | R (Intermediate/Advanced) | SQL (Intermediate) | Java (Intermediate) | C (Intermediate/Advanced) | HTML/CSS/JavaScript (Intermediate)

Modeling: Compartmental disease models, time-series forecasting (ETS, ARIMA, STL), parameter sweeps, quantile forecasts, weighted interval scoring

Data & Tools: Git/GitHub, Jupyter, RStudio, Microsoft Office Suite, Google Workspace, PRISM climate data, Quarto/R Markdown, AI tools (ChatGPT, Claude)

Research: Reproducible workflows, scientific writing, data visualization (ggplot2/tidyverse, matplotlib, Quarto/R Markdown), geospatial/climate datasets, regression analysis, experimental design

RESEARCH & PROJECT EXPERIENCE

Climate-Driven Disease Modeling — Principal Investigator | Northern Arizona University *2025–2026*

- Designed and built a spatially explicit compartmental disease model in Python (epymorph) representing connected pond networks with within-pond transmission, seasonal adult migration via distance-based movement kernels, chronic carrier adults, and environmental forcing — testing how landscape connectivity shapes ATV spread across Arizona tiger salamander metapopulations.
- Integrated daily PRISM climate data to drive temperature-dependent transmission; stepwise scenario comparisons (baseline → seasonal movement → temperature forcing) showed severe outbreaks emerge only when susceptible larvae, carrier adults, and favorable temperatures overlap seasonally.
- Compared model outputs against historical ATV prevalence records across Arizona regions; 2019 vs. 2020 differences in outbreak timing motivated the spatial framework. Code: *github.com/adk254/HURA_25-26*. Presented at OURCA Showcase, UGRADS, Undergraduate Symposium, and HURA culmination.

Influenza Forecasting — Time-Series Modeling Project *Spring 2026*

- Built and compared six probabilistic ILI% forecasting models in R (random walk, untransformed STL+ETS, log, square-root, and two Box-Cox variants) using hubverse-compatible output formats and CDC FluSight quantile forecast targets.
- Final model (adk-stletsbc, Box-Cox λ=0.15) achieved best validation weighted interval score of 0.2987 with 90% interval coverage of 0.92 — improving on the untransformed STL+ETS baseline (WIS 0.3501) and all other variants. Managed outputs and submission via Git/GitHub.

ADDITIONAL EXPERIENCE

Field Research Technician — ATV Salamander Sampling | Northern Arizona University *Summer 2025*

- Conducted fieldwork across Arizona collecting environmental swab and tissue samples from wild tiger salamander populations at multiple pond sites to test for Ambystoma tigrinum virus (ATV) prevalence.
- Navigated remote field locations, maintained sample integrity and chain-of-custody protocols, and contributed data that directly informed the spatial disease modeling work in my HURA project.

CS 345 Grader | Northern Arizona University *Spring 2025*

- Assessed student code submissions for correctness and quality; provided written feedback and flagged academic integrity concerns to faculty.